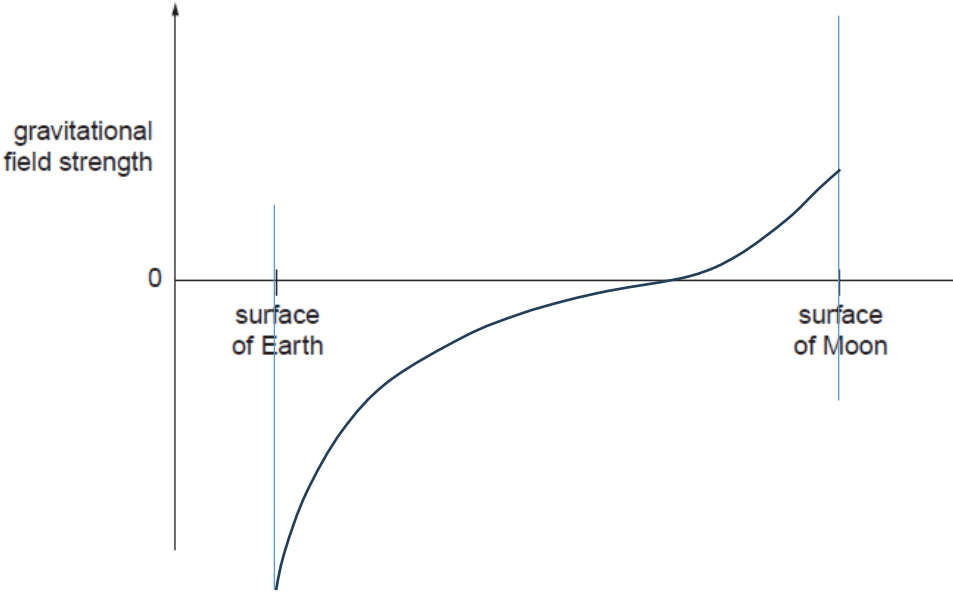


Qns	Answer	Marks
7 (a)	work done per unit mass in bringing a small test mass from infinity to that point	B1 B1
7(b)(i)	gravitational field strength of the electric field at a point is numerically equal to the potential gradient at that point. field strength = $\frac{-0.80 \times 10^7 - (-4.00 \times 10^7)}{5R - 0}$ $= \frac{3.2 \times 10^7}{5(6.4 \times 10^6)}$ $= 1.0 \text{ N kg}^{-1}$ (accept 0.9 to 1.2 N kg⁻¹)	C1 C1 A1
7(b)(ii)1	Point A. Since the total energy A intersects the potential energy graph at $x = 2R$, the space capsule would only possess gravitational potential energy and have <u>zero kinetic energy</u> .	A1 M1
7(b)(ii)2	Point(s) B, C, D. Total energy is greater than GPE. KE is not zero	A1 M1 M1
7(b)(ii)3	Point D. As the total energy D value is equal to zero, the space capsule would have enough energy to travel to <u>infinity</u> where the <u>potential energy is zero</u> .	A1 M1
7(b)(ii)4	Point B. For circular motion, $TE = -\frac{GMm}{2r} = \frac{1}{2} \left(-\frac{GMm}{r} \right) = \frac{1}{2} (GPE)$	A1 M1

7(c)(i)	 Shape $g = 0$ at <i>least</i> $\frac{2}{3}$ distance to Moon $g_E > g_M$ at surface	M1 B1 B1
7(c)(ii)	Minimum energy = $m \Delta\Phi$ $= (4.0)[-1.3 - (-62.3)] \times 10^6$ $= 2.44 \times 10^8 \text{ J}$	C1 C1 A1